Project M.A.L.L.I.

Mobile Advanced Lightweight Localization & Imaging

Overview

Project M.A.L.L.I. is a biomedical engineering initiative designed to decentralize malaria diagnostics. By combining **MobileNetV3** computer vision with low-cost smartphone microscopy, we enable high-accuracy screening in resource-limited environments.

Biomedical Context: In oncology, we look for rare malignant cells in complex tissues. In malaria, we look for intracellular parasites within a sea of Red Blood Cells (RBCs). The math remains similar: ***P(Infection | Feature_x)***.

Phase 1: Computational Core

- **Model:** MobileNetV3 (Small) for optimal speed-to-accuracy ratio.
- **Objective:** High ***Recall*** to avoid false negatives.
- **Optimization:** 8-bit Integer Quantization for sub-100ms inference on mobile devices.

Phase 2: Hardware & Optics

- **Magnification:** Target 400x-1000x effective magnification.
- **Interface:** 3D-printable smartphone cradle to maintain optical axis alignment.
- **Field-Hardening:** Digital autofocus algorithms to compensate for mechanical tolerances of cheap lenses.

Phase 3: The Roadmap

The project follows a 12-week development cycle, moving from cloud-based model training to hardware integration and finally field validation.